

RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020 – 2021 (ODD Semester)

Innovative Practice(s) Description

Name of the Faculty : Mr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Subject Code and Title : OML 751 Testing of Materials
Year and Branch : IV year Civil, EEE and ECE
Topics Covered : Unit -II: Mechanical Testing –Tensile Test
Name of the Laboratory : Strength-of-Materials lab
Website Link : <http://sm-nitk.vlabs.ac.in/exp13/index.html>
Date : 04.09.2020
No.of Students participated : 61
Assignment given if any : -
(If Yes share the uploaded
folder link)

Demonstration of Tensile Test

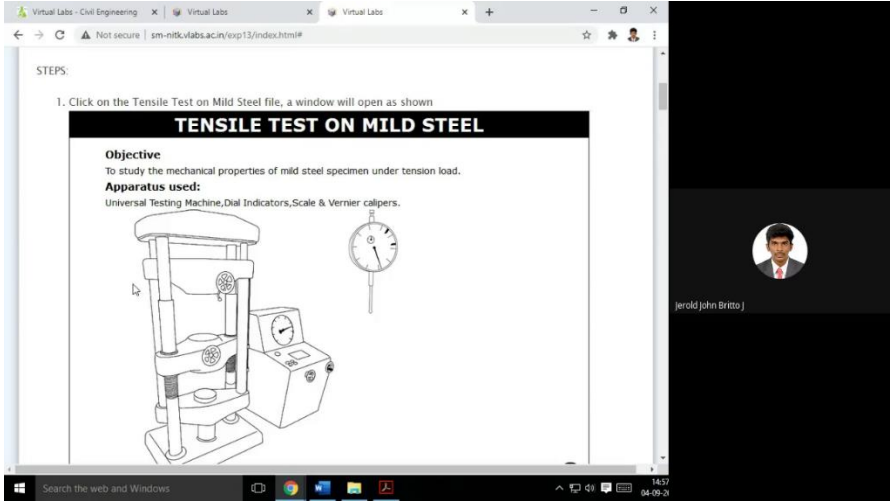
OBJECTIVE

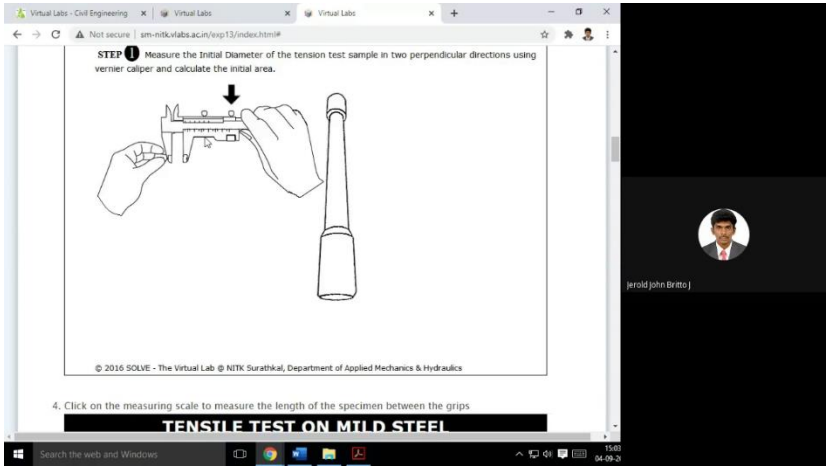
To study the mechanical properties of Mild Steel specimen under tension load.

In tension test ends of a test piece are fixed into grips connected to a straining device and to a load measuring device. The test involves straining a test piece by tensile force generally to fracture for the purpose of determining one or more of the mechanical properties.

The students can able to understand the Hooke's Law principle and the parameters like elastic limit, ultimate stress and Fracture stress values.

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO2 Analyze engineering components using various mechanical testing procedure.	2	2	2	1	1	2	2	3	1



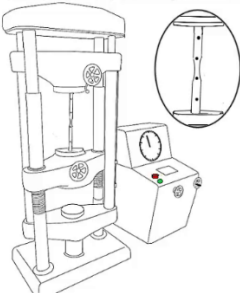


Virtual Labs - Civil Engineering Virtual Labs Virtual Labs

Not secure | sm-nitk.vlabs.ac.in/exp13/index.html#

TENSILE TEST ON MILD STEEL

STEP 6 Experiment is conducted. Load, extensometer readings are read at regular intervals of load upto the yield point. Ivory scale readings are noted till the specimen failure.



41	0	0	0	6.5
39	0	0	0	9
40	0	0	0	11
42.5	0	0	0	14
45	0	0	0	17
47.5	0	0	0	23
50	0	0	0	31
52.5	0	0	0	44
49	0	0	0	49
46	0	0	0	51
44	0	0	0	52

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